Maulana Azad National Institute of Technology, Bhopal

MidTerm Examination (March 18, 2024)

Course: B.Tech. (Semester IV)
Session: 2023-24
Subject: Mathematics - IV (MTH 241)

Branches: Chemical,EE,ME,MME
Duration: 90 Minutes
Maximum Marks: 20

Instructions: All questions are compulsory.

---

1. Find the optimal solution of the Linear Programming Problem (LPP) [Marks 3]

$$\begin{aligned} \text{Maximise} \quad & z = x_1 + x_2 \\ \text{subject to} \quad & x_1 + x_2 \le 8 \\ & 2x_1 + x_2 \le 10 \\ & x_1 \ge 0, x_2 \ge 0 \end{aligned}$$

Does there exists more than one optimal solution? Discuss your answer with reasons.

2. For the LPP given below, evaluate the optimal value of the objective function z. Also find the values of x_1, x_2 and x_3 for which the optimal value is achieved. [Marks 5]

$$\begin{aligned} \text{Maximise} \quad & z = 4x_1 + 5x_2 - 3x_3 + 50 \\ \text{subject to} \quad & x_1 + x_2 + x_3 = 10 \\ & x_1 - x_2 \ge 1 \\ & 2x_1 + 3x_2 + x_3 \le 40 \\ & x_1 \ge 0, x_2 \ge 0, x_3 \ge 0 \end{aligned}$$

3. Write the dual LPP for the following primal problem [Marks 2]

$$\begin{aligned} \text{Minimise} \quad & z = 2x_1 + 3x_2 + 4x_3 \\ \text{subject to} \quad & 2x_1 + 2x_2 + 3x_3 \le 4 \\ & 3x_1 + 4x_2 + 5x_3 \ge 5 \\ & x_1 + 2x_2 + x_3 = 7 \\ & x_1 \ge 0, x_2 \text{ is unrestricted}, x_3 \ge 0 \end{aligned}$$

4. Four engineers are available to design four projects. Engineer 2 is not competent to design the project B. Given the following time estimates needed by each engineer to design a given product, find the optimal assignment of engineers to the projects that minimises the total design time of all four projects. [Marks 4]

	Projects			
Engineers	A	B	C	D
1	12	10	10	8
2	14	Not Eligible	15	11
3	6	10	16	4
4	8	10	9	7

P.T.O.